

BAPTISTE R. JOURNAUX, Ph.D.

Assistant Professor

Department of Earth and Space Sciences, University of Washington - Seattle, USA

✉ bjournau@uw.edu • github.com/Bjournaux

Research Interests

Planetary science, experimental high pressure mineral physics, astrobiology, thermodynamic and mechanical properties water-rich minerals, petrology, interior structure and dynamic evolution of planetary bodies, robotic space exploration. Main developer of the **SeaFreeze** open-source code and **SeaFreezeGUI** for the representation of water and ice thermodynamic properties. Member of the **NASA DragonFly science team**. University of Washington expert for Space and Aeronautics.

Education

Ph.D. Planetary Sciences and Geophysics : "Petrology and rheology of high-pressure ices."	2013
<i>École Normale Supérieure de Lyon (ENS Lyon) - Université de Lyon, France</i>	
Master 2 (M.S.2) Planetary Sciences , (First class honors and highest marks)	2010
<i>Sorbonne Université (Previously Université Pierre and Marie Curie), Paris, France</i>	
Licence 3 and Master 1 (B.S.3 + M.S.1) in Physics and Chemistry of the Earth and Planets	2008 - 2009
<i>École Normale Supérieure de Lyon (ENS Lyon) - Université de Lyon, France</i>	

Professional Experiences

Assistant Professor , ESS, Astrobiology Faculty University of Washington	since 2025
Research Assistant Professor , Astrobiology Faculty, University of Washington	2023-2025
Acting Assistant Professor , University of Washington	2020 - 2023
Research associate , University of Washington	2018 - 2019
NASA Post-doctoral Program Fellow , University of Washington	2016 - 2018
Post-doctoral Fellow in experimental Glaciology , Université Grenoble Alpes, France.	2014 - 2016
<i>Mentor : Dr. M. Montagnat</i>	
Ph.D. in Mineral Physics and Thermodynamics	2010 - 2013
<i>Laboratoire de Géologie de Lyon, École Normale Supérieure de Lyon (ENS Lyon), France.</i>	

Honors and Awards

2025	Helmholtz Award from the the International Association for the Properties of Water and Steam (IAPWS)
2024	Bassett Distinguished Teaching Award, ESS department, University of Washington
2023	AGU Early Career Award in Mineral and Rock Physics
2016	NASA Postdoctoral Program Fellow
2010	French gouvernement Graduate research scholarship (Bourse Ministerielle)

Main collaborators

Anna Pakhomova (ESRF, France), Ines Collings (Natural Museum London), Tizianna Boffa-Ballaran (BGI, Germany), Steve Vance (NASA/JPL), Jacob Buffo (Dartmouth College), Morgan Cable (NASA/JPL), Matt Powell-Palm (Texas A&M), Michelle Muth (UW), Joshua Krissanssen-Totton (UW), Élodie Lesage (NASA/JPL), Gabriel Tobie (CNRS, France), Maurine Montagnat (CNRS, France).

Grants (UW Portion indicated in parenthesis)

Awarded:

Synchrotron Beamtime at ESRF, PETRA III and APS as PI (> 900h, or >\$1.8M equivalent).	2012 - 2025
NASA FINEST - PI, PhD student FI: Lauren Kafadian (\$150k for UW).	2025
NASA ROSES SSW - Community of Practice Studying Liquids in Ocean Worlds, Co-PI (\$700k).	2024
NASA ROSES SSW - Ocean-Ice shell Dynamic Evolution, Co-PI (\$350k).	2024
NASA ROSES PSIE -Preparatory Science Investigations for Europa, Science-PI (\$450k).	2024
NASA ROSES COVID Augmentation for Planetary Solution Chemistry (\$70k).	2020

NSF The Chemistry of Aqueous Carbonic Fluids in Subduction, Co-I. (\$393k)	2018
NASA ROSES-SSW Planetary Solution Chemistry, Co-I. (\$370k)	2017
NASA Postdoctoral Fellowship , Sole PI (\$200k)	2016
French gouvernement Ph.D. fellowship (3 years of salary \$120k)	2010

Pending:

NASA PDAR: Effects of high pressure on the equilibria of brines relevant to icy bodies, PI (\$450k).	2024
NASA ICAR: Habitable Geophysics, Science-PI (\$1.2M).	2024

Science Team member

IAPWS working group on physical properties of high pressure ice polymorphs	since 2024
NASA Preparatory Science Investigations for Europa-Clipper <i>Team PI: Steve Vance, Jet Propulsion Laboratory, Pasadena, CA, USA</i>	since 2024
NASA DragonFly mission team (Collaborator) <i>Team PI: Elizabeth "Zibi" Turtle, Johns Hopkins University Applied Physics Laboratory, MD, USA</i>	since 2021
NASA Astrobiology Institute - Habitability of hydrocarbon worlds, Titan and Beyond (Co-I) <i>Team PI: Rosaly Lopes, Jet Propulsion Laboratory, Pasadena, CA, USA</i>	since 2018

Selected Invited Seminars/Lectures since 2019

Upcoming

○ Gordon Research Conference: Rock Deformation, Lewiston, MA, USA	Aug. 2026
○ WaterX conference, La Maddalena, Sardinia, Italy	May 2026

Previous

○ IPGP, Planetary science team seminar, Paris, France	Oct. 2025
○ University of Washington, Astrobiology seminar, WA, USA	Oct. 2025
○ Brown University, Planetary science seminar, RI, USA	April. 2025
○ MIT, Planetary science seminar, MA, USA	April. 2025
○ University of Washington, Chemistry department, seminar, WA, USA	Mar. 2025
○ Cornell University, Carl Sagan Institute seminar, NY, USA	Feb. 2025
○ University of Washington, ESS department, seminar, WA, USA	Feb. 2025
○ Plenary at Gordon Research Conference: Water and Aqueous Solutions, Holderness, NH, USA	July. 2024
○ NASA DragonFly science team annual meeting, JHAPL, Laurel, Maryland, USA	Feb. 2024
○ NASA JPL science seminar, Pasadena, CA, USA.	Sep. 2023
○ ESA JUICE MAJIS science team seminar serie (online)	Jul. 2023
○ NASA Europa Clipper surface science team seminar serie (online)	May 2023
○ NASA DragonFly science team seminar serie (online).	Mar. 2023
○ Western Washington University, Bellingham, WA, USA	Oct. 2022
○ NASA Europa Clipper Lecture Series (online)	Jun. 2022
○ NASA Astrobiology Institute Titan Seminar Series, University of Hawaii, Honolulu, USA.	May. 2022
○ Carnegie Institution for Science - Earth and Planets Laboratory, (online)	Jan. 2022
○ University of Nantes, France, (online)	Jun. 2021
○ UT-Austin, Jackson School of Geosciences (online)	Apr. 2021
○ NASA JPL ICE research group Seminar, (online)	Jan. 2021
○ NASA Astrobiology Institute Titan Seminar Series, Caltech, Pasadena, USA,	Jan. 2020
○ Lawrence Livermore National Laboratory, High Energy Density Science.	Nov. 2019
○ UC-Santa Cruz - Earth & Planetary Sciences, Whole Earth Seminar series.	Oct. 2019
○ ETH Zurich, Geosciences department.	Sep. 2019
○ University of Washington Earth and Space Science Seminar, Seattle USA,	Apr. 2019
○ Arizona State University, School of Earth and Space Exploration	Apr. 2019
○ ESRF-BSD Extreme Conditions Topical Bridge seminar, ESRF, Grenoble, France	Nov. 2018
○ Sorbonne University, IMPMC/LULI labs seminars, Paris, France	Nov. 2018
○ Earth and Life Science Institute, Tokyo Institute of Technology, Tokyo, Japan.	Jan. 2018
○ Jet Propulsion Laboratory / Caltech seminar and visiting scientist, Pasadena.	Oct. 2017
○ University of Nevada, Las Vegas, Physics department seminar.	Sept. 2017
○ Bayerisches Geoinstitut, Bayreuth, Germany.	Aug. 2016

Past and present students and postdocs mentored

Underlined at UW; [‡]Co-advised

Postdoc Mentee:

- Kasia Skrzyńska[‡] (since 2025), co-advised with A. Pakhomova at the European Synchrotron Radiation Facility, Grenoble, France.
- Fabian Klenner[‡] (2023-2025), co-advised with D. Catling at UW ESS; Now assistant professor at UC Riverside

Graduate Student Mentees

Current:

- Ula Jones, PhD student since fall 2023, (Expected Fall 2028); Topic : Equation of states of high pressure ice polymorphs VII-X, water-rich exoplanet interior structure and dynamic evolution. *Awards : CENV travel, ESS Graduate student support fund, ARCS Foundation Directed Fellowship, Provost Award.*
- Haskelle White Gianella[‡], PhD student since fall 2023, (Expected Fall 2028); Topic : Constraining Venus past habitability from modeling and experimental mineralogy. (Co-advised with Dr. Joshua Krissansen-Totton at UW ESS). *Awards : NSF GRFP, ESS research award, UW-Student Technology Fee.*
- Lauren Kafadarian, PhD student since winter 2025, (Expected Fall 2029); Topic : Ultrasonic and thermodynamics properties of icy moons oceans at elevated pressures; *Awards : NASA FINESST, ESS research award.*
- Ifigeneia Rousouli, PhD student since fall 2025, (Expected Spring 2030); Topic : Mineralogy and thermodynamic properties of mixed organic/salt systems at Europa and Titan conditions
- Scott Chang[‡], PhD student since fall 2025, (Expected Spring 2030); Topic : Synergistic geophysical analysis of icy worlds spacecraft data using machine learning. (Co-advised with Dr. Steve Vance at NASA JPL) *Awards : Provost Award, Washington Space Grant - SURP*
- Shradha Ravikumar[‡] (Harvard University), PhD student since fall 2025, (Expected Spring 2030); Topic : High pressure ices experimental rheology. (Co-advised with Dr. Rebecca Fischer at Harvard) *Awards : NSF GRFP Honorable mention.*

Former:

- Dr. Christopher Woodburn, PhD. (2023-2025); PhD Thesis: Advancement in high pressure chemistry: Informing biosignature detection and aqueous salt equilibria. (Transfer from MSE)
- Penny Espinoza. (2018-2020), Topic: Equation of states representation with piecewise polynomial.

Undergraduate Mentees (^hHonors)

Current:

Tristen Arias^h (UW Physics) (since 2025), *Award: Mary Gates Research Scholarship, LPSC 2026 Poster presentation;*
Catie Zeiger^h (UW ESS/pre-Med) (since 2024)

Former:

Elizabeth Pawelka^h (UW Physics, 2023–2025)
Now PhD student, Rice University

Arian Zarriz (Texas A&M, 2022–2025)
Now PhD student, Texas A&M

Jack Rosenbloom (UW CSE, 2023–2024)
Bing Hong Chua (NASA-JPL, 2023)
Now PhD student, UCLA

Jillian Florent^h (UW ESS, 2023)

Abby Sylvester^h (UW ESS, 2023)

Evan Carnahan^h (UT Austin, 2021-2022)

Karlee Taylor (Macalester College, 2021–2022)
Now PhD student at UW-ESS

Wei Dai^h (UW CSE, 2020)

Erica Clinton^h (UW Chemistry, 2020–2021)
Now PhD student at ETH Zurich and Paul Scherrer Institut

Lane Lindblom^h (UW ESS, 2019)

Nathan Reinsdorf^h (UW ESS, 2018)

Dr. Jason Ott^h (UW ESS, 2016–2019)
Now Postdoctoral fellow at Rice University

Fred Bradley (2017)

Léa Gest (2014–2015)

Dr. Jean Alexis Hernandez (2012)
Now Beamline scientist, European Synchrotron

Graduate Committees Served on:

Anna Grace Ulses (UW ESS, since 2023)
Andrew Shumway (UW ESS, 2022-2025)

George Kanaan (UW Oceanography, 2022-2025)
Chaman Gupta (UW MSE, 2023-2025)

Other Graduate Students mentored:

Andrew Nuhaily, PhD Student, UW- MSE, (since 2025)
Arian Zarriz, PhD Student, Texas A&M, (since 2025)
Nicole Sutherland, PhD Student, NHM, UK (since 2025)
Julia Miller, PhD Candidate, Brown (since 2024)
Catherine Psarakis, PhD Candidate, UCLA, (since 2022)

Dr. Andrea Bryant, University of Chicago, (2021-2024)
Now Postdoc at Brown University, USA
Dr. Pauline Leveque, Univ. of Nantes, France (2021-2024)
Now Research Scientist at Institute of Science Tokyo, Japan

Courses Taught

ESS 102 - Introduction to Space Travel and Planetary Exploration	since 2019 (16 quarters)
ESS 490/590 - Graduate Seminar on Experimental techniques in Planetary and geosciences	2025
ESS 490/590 - Graduate Seminar on Planetary Interiors	2024
AB 501 - Graduate Seminar on Astrobiology (lead: Prof. Rory Barnes in astronomy)	2022, 2024
ESS 202 - Earthquakes	2019
TA in France (190h) in geosciences, crystallography, planetary sciences, fieldwork	2010-2013

Leadership and Services

National and International Service:

- IAPWS working group lead on ices polymorphs properties since 2025
- AGU Mineral and Rock Physics Program committees since 2024
- Cryobiology Research membership committee since 2023
- AGU Fall meeting Primary Session convener 2018, 2019, 2021, 2022, 2024
- Astrobiology Science Conference (AbSciCon) Primary Session convener 2017, 2019, 2022, 2024, 2026
- NASA grant proposal review panelist
- Peer review : Nature Astronomy, Nature Communications, JGR Planets, Icarus, ACS Earth and Space Chemistry, ACS Physical Review X, Proceedings of the Royal Society A, The Astrophysical Journal, Cryobiology.
- Glaciology fieldwork for GLACIOCLIM and WGMS (INSU/IPEV/OSUG/LGGE/WGMS) 2015.
- Early career scientist (ECS) committee for the EGU-Planetary Science section. 2014-2017
- Organizing committee for the 1st Astrobiology Graduates in Europe Conference, Edinburgh, UK 2014

University/Department Service

- ESS Graduate Admission Committee 2023,2025, 2026
- ESS Laboratory Safety Committee since fall 2025
- ESS Renovation Committee 2024
- ESS Graduate prelim exams 2024
- UW-ESS open house solar observation activity 2024, 2025
- UW-ESS graduate student trip for the 2024 total solar eclipse, produced text, image and video content for ESS and the college of the Environment communication 2024
- UW Astrobiology seminar program committee spring 2017

Public Outreach

Press Interviews

PBS/Associated Press, "Saturn's moon Titan may not have a buried ocean as suspected, study suggests", Sara Hashemi, Dec. 17 2025

Smithsonian Magazine, "Saturn's Moon Titan May Not Have an Underground Ocean After All", Marcia Dunn, Dec. 17 2025

Gizmodo, "Scientists Thought Saturn's Moon Titan Hid a Secret Ocean. They Were Wrong", Elyn Lapointe, Dec. 17 2025

Sciences et Avenir (Largest science magazine for the general public in France), "Sans doute pas d'océan sous la surface de Titan, la lune de Saturne", Joel Ignasse, Dec. 23 2025

Le Figaro (Second-largest national newspaper in France), "Une vaste couche de glace «chaude», idéale pour le développement de la vie, cachée dans les entrailles de Titan", Vincent Bordenave, Dec. 17 2025

UW News, "Scientists rule out a hidden ocean on Titan, but there's still hope for discovering life under the ice", Gillian Dohrn, Dec. 17 2025

GeekWire, " Saturn's biggest moon might not have a global ocean — but the search for life isn't over", Alan Boyle, Dec. 17 2025

Salon, "Why freezing cold worlds could be our best bet for alien life in the solar system", M. Rozsa, Jan. 4 2025

UW News, "In the Field: UW researchers traveling to capture total solar eclipse", Apr. 4 2024

New York Times, "Take Water. Add Sodium Chloride. Chill and Squeeze Into Salty Ices.", K. Chang, Feb. 25, 2023

UW News, "Newly discovered form of salty ice could exist on surface of extraterrestrial moons", Hannah Hickey, Feb. 21 2023

CNN, "New type of salty ice may exist on extraterrestrial ocean moons.", A. Strickland, Feb. 20, 2023

UW News, "Experiments measure freezing point of extraterrestrial oceans to aid search for life", Hannah Hickey, May. 3 2022

Podcast/ Radio Interviews

KUOW, Exploring the moons of our solar system, TBD, 2026

NewScientist Dead Planets Society podcast, Episode 2: Punch A Hole in a Planet, Jul. 31, 2023

Exocast-45b: Special Guest Dr Baptiste Journaux, Jun. 2, 2020

Strange New Worlds Episode 76 , Aug. 11, 2019

Podcast Science (in French), Episode 389: Astrobiology and icy moons with Dr Baptiste Journaux, Jun. 11, 2019

Outreach events

Solar observation for the ESS department Open House 2024 & 2025

2nd grade Class intervention on Solar observations Bertschi School, 2024

6th grade Class intervention on Mineralogy Washington Middle School, 2023

Seattle Astro on Tap Presentation on Planet interiors, 2019

Publications

Total number: 34

First/senior-authored: 12

h-index: Google Scholar, 16; Scopus, 15 (as of December 2025)

ORCID 0000-0002-0957-3177

* When corresponding author

§ Mentee

† Primary advisor

‡ Co-advising

In preparation (4)

4. (‡) A. S. Bryant[§], **B. Journaux**, M. P. Panning, A. G. Marusiak, F. Petricca, "Seismic and Thermal Profiles of High-Pressure Ice in Titan's Hydrosphere", in prep.
3. (†) T. Arrias[§], **B. Journaux***, "Subsurface thermal state model of Lunar permanent shadowed regions", in prep.
2. (†) U. Jones[§], J. M. Brown, **B. Journaux***, "Thermodynamic properties at solution ices-equilibria in the H₂O-NaCl system, implication for icy worlds and astrobiology", in prep.
1. (†) C. Woodburn[§], M. J. Pasterski, K. M. Seaton, C. Gupta, Z. Feng, T. Weidner, X. Li, S. Chandra, B. L. Henderson, D. A. Kaplan, F. van Amerom, M. E. Castillo, J. Graham, R. M. Danell, W. B. Brinckerhoff, A. Grubisic, M. G. Trainer, M. L. Cable, **B. Journaux***, "Inhibition of Ice Recrystallization by a Peptide at Gigapascal Pressures: A Functional and Detectable Biosignature for Icy Moons", in prep.

Under-review (7)

7. (†) U. Jones[§], J.M. Brown, **B. Journaux**, "Meta-analytical Gibbs energy equation of state of ice VII-X", Under review in JGR-Planets.
6. (†) H. T. White-Gianella[§], V. Chevrier, **B. Journaux**, "Thermodynamic Modeling of Alterations During Climate Transition Reveals Minerals Indicative of Past Temperate Conditions on Venus" Under review in Icarus.
5. (†) C. Woodburn[§], S. Vance, J.M. Brown, **B. Journaux***, "On the Melting Curves of Water at Pressure and in the Presence of Salts Expected in Icy Worlds", under review in PNAS
4. D. R. Rice, C. Huang, R. Royer, M. Daspute, K. Mittal, **B. Journaux**, J. H. Steffen, A. Vazan, "Magrathea v2: A planetary interior modeling platform in C++", Under review in JOSS
BRJ contribution: methodology, review and editing.
3. (‡) K. Skrzyńska[§], E. Le Menn, O. Bollengier, **B. Journaux**, A. Pantousas, M. Mezouar, G. Tobie and A. Pakhomova¹, "Overcoming hydrophobicity: A maximally filled methane clathrate hydrate reservoir under planetary conditions", Under review in Nature Communications
2. J. M. Brown, O. Bollengier, U. Jones, S.D. Vance, **B. Journaux***, "Thermodynamics of NaCl(aq) from 230 to 2000 K and from 1 bar to 10,000 MPa: Applications to Earth and Other Ocean Worlds", Under review in Astrochimica and Geochemica Acta
1. M. Cantine, **B. Journaux**, H. Griffiths, E. Mitchell, C. Ostrander, H. Cothren, U. Jones. "The Avalon assemblage as astrobiological analog for icy ocean worlds", Under review in Astrobiology

Peer-reviewed (34)

34. I. Collings, A. Pakhomova, **B. Journaux**, T. Boffa Ballaran, A. Kurnosov, J. M. Brown, G. Garbarino, J. Smith, "Polymorphism of magnesium chloride decahydrate under high pressures and low temperatures found in icy moons". (2026) ACS Earth Space Chem., 10, 2, 434–440
33. F. Petricca, S. D. Vance, M. Parisi, D. Buccino, G. Cascioli, J. Castillo-Rogez, B. G. Downey, F. Nimmo, G. Tobie, **B. Journaux**, A. Magnanini, U. Jones, M. Panning, A. Bagheri, A. Genova, J. Lunine, "Titan's Strong Tidal Dissipation Revealed by Cassini Data Precludes a Subsurface Ocean" (2025) Nature, 648, pp. 556–561.
32. D. Apai, R. Barnes, T. Lichtenberg, M. Murphy, N. Tuchow, R. Ferriere, K. Wagner, A. Affholder, R. Malhotra, **B. Journaux**, A. Vazan, R. Ramirez, N. Y. Kiang, A. Mendez, S. R. Kane, V. H. Klawender, and NExSS Quantitative Habitability Science Working Group, "A Terminology and Quantitative Framework for Assessing the Habitability of Solar System and Extraterrestrial Worlds" (2025) Planetary Science Journal, 6, 165.
31. Anna Pakhomova, **B. Journaux***, A. Kurnosov, T. Boffa Ballaran, G. Tobie, M. Hanfland, "Methanol storage in high-pressure clathrate hydrates as a prolonged source of methane in large ocean worlds" (2025) Earth and Planetary Science Letters, 463, 36–47.

30. (‡) J. N. Ott[§], C. B. Condit, M. Pec, **B. Journaux**, "Dislocation creep of glaucophane in mafic blueschists during subduction: weighted Burgers vector analysis from the Catalina Schist" (2025) *Geology* 53 (6): 529–534.
29. (‡) F. Klenner[§], L. Fifer, **B. Journaux**, A. Bravenec, A. M. Leal, S. Vance, D. Catling, "Supercooling, Glass Formation and Mineral Assemblages Upon Freezing of Salty Ice Grains from Enceladus's ocean" (2025) *Planetary Science Journal*, 6,3, 65.
28. (‡) A. Zarriz[§], **B. Journaux***, M. Powell-Palm, "On the equilibrium limit of liquid stability in pressurized aqueous systems" (2024) *Nature Communications*, 15, 10666.
27. (‡) C. Psarakis[§], T. Fidelis, K. Chin, **B. Journaux**, A. Kavner, P. Sarker, M. J. Styczinski, S. D. Vance, T. Wei, "Electrical Conductivity of Subsurface Ocean Analogue Solutions from Molecular Dynamics Simulations" (2024) *ACS Earth Space Chem.*, 8, 6, 1146–1153.
26. L. Ojha, J. Buffo, **B. Journaux**, "Thermophysical arguments against basal melting in the south polar region of Mars." (2024) *Icarus*, 407(1), 115772.
25. (‡) B. Hong Chua[§], E. Gloesener, M. Choukroun, T. H. Vu, M. Melwani Daswani, **B. Journaux**, M. J. Styczinski, S. D. Vance, "Low-Temperature Specific Heat Capacity of Water-Ammonia Mixtures Down to the Eutectic" (2023), *ACS Earth Space Chem.*, 7, 10, 1971–1979 .
24. **B. Journaux***, A. Pakhomova, I. E. Collings, S. Petitgirard, T. Boffa Ballaran, J. M. Brown, S. D. Vance, D. Huang, S. Chariton, V.B. Prakapenka, J. Ott, D. Comboni, M. Hanfland, "On the identification of hyperhydrated sodium chloride hydrates, stable at icy moon surface conditions." (2023). *PNAS*, 120 (9) e2217125120.
23. L. Ojha, B. Troncone, J. Buffo, **B. Journaux**, G. McDonald, "Liquid Water on Cold Exo-Earths via Basal Melting of Ice Sheets" (2022), *Nature Communications*, 13, 7521.
22. A. G. Marusiak, S. Vance, M. P. Panning, A. S. Bryant, M. A. Hesse, E. Carnahan, **B. Journaux** "The Effects of Methane Clathrates on the Thermal and Seismic Profile of Titan's Icy Lithosphere" (2022) *Planetary Science Journal*, 3, 167.
21. **B. Journaux***, "Salty ice and the dilemma of ocean exoplanet habitability." (2022) *Nature Communication* 13, 3304.
20. (‡) E. Carnahan[§], S. Vance, M. Hesse, **B. Journaux**, C. Sotin, "Dynamics of mixed clathrate-ice shells on icy ocean worlds". (2022) *Geophysical Research Letters*, 49(8).
19. (‡) B. Chang[§], A. N. Consiglio, D. Lilley, B. Rubinsky, **B. Journaux**, M. J. Powell-Palm, "On the pressure dependence of salty aqueous eutectics". (2022) *Cell Reports Physical Science*, 3, 100856.
18. A. Pakhomova, I. Collings, **B. Journaux**, S. Petitgirard, T. Boffa Ballaran, D. Huang, J. Ott, A. Kurnosov, M. Hanfland, G. Garbarino, D. Combini, "Host-guest hydrogen bonding in high-pressure acetone clathrate hydrates: in situ single-crystal X-ray diffraction study", (2022) *The Journal of Physical Chemistry Letters*, 13, 1833–1838.
17. A. Marusiak, S. Vance, M.P. Panning, M. Behounkova, P.K. Byrne, G. Choblet, M. Melwani Daswani, K. Hughson, **B. Journaux**, E. Kite, A.H. Lobo, K. Pleiner Sladkova, K. M. Soderlund, W. Song, A. Soucek, G. Steinbrugge, A.F. Thompson, S. Wang, N. S. Wolfenbarger, "Exploration of Icy Ocean Worlds using Geophysical Approaches". (2021) *The Planetary Science Journal*, 2:150 (14pp).
16. S. MacKenzie, S. Birch, S. Horst, C. Sotin, E. Barth, J. M. Lora, M. G. Trainer, P. Corlies, M. J. Malaska, E. Sciamma-O'Brien, A. E. Thelen, E. Turtle, J. Radebaugh, J. Hanley, A. Solomonidou, C. Newman, L. Regoli, S. Rodriguez, B. Seignovet, A. G. Hayes, **B. Journaux**, J. Steckloff, D. Nna-Mvondo, T. Cornet, M. Palmer, R. M.C. Lopes, S. Vinatier, R. Lorenz, C. Nixon, E. Czapinski, J. W. Barnes, E. Sittler, and A. Coates, "Titan: Earth-like on the Outside, Ocean World on the Inside". (2021) *The Planetary Science Journal*, 2:112 (13pp).
15. S. Vance, **B. Journaux**, M. Hesse, G. Steinbrügge, "The Salty Secrets of Icy Ocean Worlds". (2021), *Journal of Geophysical Research - Planets*, 2020JE006736RR
14. **B. Journaux***, J.M. Brown, A. Pakhomova, I. Collings, T. Boffa-Ballaran, S. Petitgirard, J. Ott, "Holistic approach for studying planetary hydrospheres: Gibbs representation of ices thermodynamics, elasticity and the water phase diagram to 2300 MPa." (2020), *Journal of Geophysical Research - Planets*, 125(1), e2019JE006176
13. J.M. Brown, **B. Journaux***, "Local Basis Function Equations of State for High Pressure Ice VII and X". (2020) , *Minerals*, 10(2), 92, Special Issue : Mineral Physics, In Memory of Orson Anderson.
12. Ruth-Sophie Taubner, K. Olsson-Francis, S. Vance, A. Antunes, E. Camprubi Casas, J.P. de Vera, J. Goodman, **B. Journaux**, F. Klenner, L. Noack, F. Postberg, E. Rabbow, N. Ramkissoon, T. Ruckriemen-Bez, J. Saur, Y. Sekine, K. Soderlund, "Experimental and simulation efforts in the Astrobiological exploration of Exooceans". (2020) *Space Sci. Rev.* 216, 9.
11. **B. Journaux*** K. Kalousova C. Sotin, G. Tobie, S. Vance, J. Saur, O. Bollengier, L. Noack, T. Ruckriemen-Bez, T. Van Hoolst. K. M. Soderlund, J. M. Brown, "Large ocean worlds with high-pressure ices". (2020) *Space Sci. Rev.* 216, 7.
10. S. Jarmak, E. Leonard, L. Schurmeier, A. Akins, E. Dahl, D. R. Cremons, S. Cofield, A. Curtis, C. Dong, E. T. Dunham, **B. Journaux**, D. Murakami, W. Ng, M. Piquette, A. Pradeepkumar Girija, K. Rink, N. Stein, N. Tallarida, M. Telus, L. Lowes, C. Budney, K. L. Mitchell, "QUEST: A New Frontiers Uranus Orbiter Mission Concept Study". (2020) *Acta Astronautica*, 170, 6-26.

9. **B. Journaux*** T. Chauve, M. Montagnat, A. Tommasi, D. Mainprice, F. Barou, L. Gest, "Microstructure and texture evolution in polycrystalline ice during hot torsion. Impact of intragranular strain and recrystallization processes." (2019) *The Cryosphere*, 13, 1-17.
8. E. H. Abramson, O. Bollengier, J. M. Brown, **B. Journaux**, W. Kaminsky, A. Pakhomova, "A new high-pressure carbonic acid hydrate structure determined in-situ." (2018) *American Mineralogist* 103(9), 1468-1472.
7. T. Chauve, M. Montagnat, S. Piazzollo, **B. Journaux**, J. Wheeler, F. Barou, D. Mainprice, A. Tommasi: "Non-basal dislocations should be accounted for in simulating ice mass flow" (2017) *Earth and Planetary Science Letters*, 473, 247-255.
6. **B. Journaux***, I. Daniel, S. Petitgirard, J-P. Perillat, H. Cardon, "Salt partitioning between water and high-pressure ices. Implication for the dynamics and habitability of icy moons and water-rich planetary bodies". (2017) *Earth and Planetary Science Letters*, 463, 36-47.
5. S. Petitgirard, W. Malfait, **B. Journaux**, I. Collings, I. Kantor, A. Kurnosov, T. Dane, M. Burghammer, M. Hanfland, D. Rubie, "SiO₂ glass density to lower-mantle pressures". (2017) *Phys. Rev. Lett.* 119, 215701
4. L. Noack, D. Honing, A. Rivoldini, C. Heistracher, N. Zimov, **B. Journaux**, H. Lammer, T. Van Hoolst, J.H. Bredehoft, "Water-rich planets: how habitable is a water layer deeper than on Earth?" (2016) *Icarus*, 277, 215-23
3. **B. Journaux***, R. Caracas, P. Carrez, K. Gouriet, P. Cordier, I. Daniel, "Elasticity and dislocations in ice X under pressure" (2014) *Physics of the Earth and Planetary Interiors*, 236, 10-15.
2. L. Bezacier, **B. Journaux**, J-P. Perillat, H. Cardon, M. Hanfland, I. Daniel, "Equations of state of ice VI and ice VII at high temperature and high pressure" (2014) *The Journal of Chemical Physics*, 141, 104505.
1. **B. Journaux***, I. Daniel, R. Caracas, G. Montagnac, H. Cardon, "Influence of NaCl on ice VI and ice VII melting curves up to 6 GPa, implications for large icy moons." (2013) *Icarus* 226, 355-363.

Book Chapter

- S. Vance, A. Crósta, M. Melwani-Daswani, S. A. Fagents, **B. Journaux** C. D. Neishe, C. Jamieson, M. Choukroun, "Chapter 13 : Exchange processes between surface, atmosphere, and interior" (2024), in "Titan After Cassini-Huygens", Elsevier (in Press)
- B. Journaux*** K. Kalousova C. Sotin, G. Tobie, S. Vance, J. Saur, O. Bollengier, L. Noack, T. Ruckriemen-Bez, T. Van Hoolst. K. M. Soderlund, J. M. Brown, "Large ocean worlds with high-pressure ices". (2020), "in Ocean Worlds", ISSI-Springer publication (Same as Space science review publication)
- Ruth-Sophie Taubner, K. Olsson-Francis, S. Vance, A. Antunes, E. Camprubi Casas, J.P. de Vera, J. Goodman, **B. Journaux**, F. Klenner, L. Noack, F. Postberg, E. Rabbow, N. Ramkissoon, T. Ruckriemen-Bez, J. Saur, Y. Sekine, K. Soderlund, "Experimental and simulation efforts in the Astrobiological exploration of Exooceans" (2020) in "Ocean Worlds", ISSI-Springer publication. (Same as Space science review publication)
- S. Vance, T. Loerting, J. Stern, M. Kropf, **B. Journaux**, C. Jamieson, M. L. Cable, O. Bollengier, "Chapter 3 : Solids and fluids at low temperatures, in Low Temperature Materials and Mechanisms" (2016) (ed. Y. Bar-Cohen), CRC-press.

Conference Communications

§ Mentee

† Primary advisor

‡ Co-advising

Selected Oral Communications

- (‡) S. Chang[§], **B. Journaux**, S.D. Vance, F. Petricca, "Neural Network–Based Geophysical Inference of Europa's Ocean Composition." **AbSciCon 2026**, Madison, WI, USA, May 2026.
- (‡) H. T. White-Gianella[§], J. Krissansen-Totton, **B. Journaux**, J. Weber, "Crustal oxidation experiments to test the plausibility of oxygen biosignature false positives." **AbSciCon 2026**, Madison, WI, USA, May 2026.
- (‡) H. T. White-Gianella[§], J. Krissansen-Totton, **B. Journaux**, J. Weber, "Leveraging crustal oxidation experiments to assess the plausibility of liquid water on ancient Venus.", **Organization for Venus Early-Career Networking (OVEN) Early Career Seminar**, Oct. 2025
- (Invited plenary) **B. Journaux**, "Planetary science : the next frontier in water and aqueous solutions thermodynamic properties " **IAPWS Meeting 2025**, Helsinki, Finland, June 2025.
- B. Journaux**, A. Zarriz, M. Powell-Palm, "Cenotectic: A New Thermodynamic Definition of the Universal Limit of Liquid Stability and Its Implications for Icy World Structure and Habitability Near and Far" **Lunar and Planetary Science Conference**, Woodlands, TX, Mar. 2025.

- B. Journaux, A. Zarriz, M. Powell-Palm, "Eutectics under pressures: definition of a universal limit of liquid stability" European Planetary Science Conference, Berlin, Germany, Sept. 2024.**
- (Invited plenary) **B. Journaux, "A New Era of High Pressure Experiments and Numerical Representations of Aqueous Solutions and Ices Thermodynamics Motivated by Outer Solar System Exploration" Gordon Research Conference: Water and Aqueous Solutions, Holderness, NH, July 2024.**
- (†) **U. Jones[§], B. Journaux, J. M. Brown, "Examining the Equilibrium Thermodynamics of High-pressure Ice Polymorphs and Aqueous NaCl" 22nd Symposium on Thermophysical Properties and 18th International Conference on the Properties of Water and Steam, Boulder, CO, June 2024.**
- (Invited) **B. Journaux, J. M. Brown, A. Pakhomova, I. Collings, S. Petitgirard, E. Clinton, T. Boffa Ballaran and S. D. Vance, "Water Ice Polymorphs Gibbs Energy Local Basis Functions up to 2300 MPa: Reconciling Water Thermodynamics with the Water Phase Diagram" 22nd Symposium on Thermophysical Properties and 18th International Conference on the Properties of Water and Steam, Boulder, CO, June 2024.**
- B. Journaux, A. Pakhomova, I. Collings, S. Petitgirard, T. Boffa Ballaran, J. M. Brown, S. D. Vance, S. Chariton, Vitali Prakapenka and Dongyang Huang, "New Class of Stable NaCl Hydrates: On the Role of Pressure in Hydration and Ionic Dissociation in Hydrogen Bonded Solids" 22nd Symposium on Thermophysical Properties and 18th International Conference on the Properties of Water and Steam, Boulder, CO, June 2024.**
- (Invited) **B. Journaux, A. Pakhomova, M. Powell-Palm, I. Collings, S. Petitgirard, J. M. Brown, S. D. Vance, and T. Boffa Ballaran, "Salty ices for exotic oceanic cocktails: Icy worlds petrological diversity and what it can tell us about their ocean and habitability." 209-01 AbSciCon 2024, Providence, May. 2024.**
- (Invited) **B. Journaux, "300 leagues under the extraterrestrial seas: exploring the thermodynamics of Galilean moons hydrospheres using SeaFreeze" Origins and Habitability of the Galilean Moons, Marseille, France, Oct. 2023.**
- (Invited plenary) **B. Journaux, "Astro-cryobiology: Search for extraterrestrial life in the cold oceans of our solar system." CRYO2023, Mineapolis, July 2023.**
- B. Journaux, A. Pakhomova, I. E. Collings, S. Petitgirard, T. Boffa Balaran, J. M. Brown, S. D. Vance, D. Huang, S. Chariton, V.B. Prakapenka, J. Ott, D. Comboni, M. Hanfland, , "Sodium Chloride hydrates with high hydration numbers, stable at icy ocean world surface and interior conditions.", 2839, LPSC 2023, The Woodlands, TX, March. 2023.**
- B. Journaux, J. M. Brown, E. Abramson , S. Vance "Thermodynamic constraints on Titan's water-ammonia ocean habitability.", 115-06 AbSciCon 2022, Atlanta, May. 2022.**
- B. Journaux, J. M. Brown, E. Abramson , S. Vance "The thermodynamics of ammonia rich planetary oceans: Equation of State for water-ammonia and water-ammonia-salt solutions from surface to abyssal conditions.", P44A-08 AGU Fall Meeting, New Orleans, Dec. 2021.**
- B. Journaux, J. M. Brown, P. Espinoza, S. Vance, O. Bollengier, E. Abramson, A. Pakhomova, I. Collings, S. Petitgirard. "A Deep Dive into Planetary Abysses: Effects of Pressure, Phase Equilibria and Chemistry on the Structure and Habitability of Icy-Ocean Worlds", P075-05 AGU Fall Meeting, Dec. 2020.**
- B. Journaux "SeaFreeze, a new framework to represent the thermodynamic of water and the water phase diagram at extreme conditions" COMPRESS Annual Meeting 2020, Aug. 2020.**
- B. Journaux, J. M. Brown, P. Espinoza, O. Bollengier, E. Abramson, A. Pakhomova, I. Collings, S. Petitgirard., "SeaFreeze: : A holistic representation of thermodynamics and elasticity of the water phase diagram to 2300 MPa.", P52B-05, AGU Fall Meeting, San Francisco, Dec. 2019**
- (Presenting) **A. Pakhomova, B. Journaux, I. Collins, S. Petitgirard, T. Boffa-Ballaran, S. Vance, J. M. Brown, "New stable NaCl and Na₂SO₄ hydrates discovered at icy moons hydrosphere conditions.", AbSciCon 2019, Seattle, June 2019**
- (Presenting) **J.M. Brown, O. Bollengier, B. Journaux, E. Abramson, S. Vance, "Contributions of High-Pressure Chemistry to Ocean Worlds Habitability Analysis.", AbSciCon 2019, Seattle, June 2019**
- B. Journaux, J.M. Brown, A. Pakhomova, S. Petitgirard, I. Collings, O. Bollengier, T. Boffa Ballaran, J. Ott , "Comprehensive equations of state for high pressure ices and aqueous brines for accurate thermodynamic properties predictions", MR13A-06, AGU Fall Meeting, Washington DC., December 2018**
- (Presenting) **J.M. Brown, O. Bollengier, B. Journaux, S. Vance, G. Tobie, "Impact of High-Pressure Chemistry on Planetary Habitable Zones", MR22A-04, AGU Fall Meeting, Washington DC., December 2018**
- (Invited) **B. Journaux, J.M. Brown, E. Abramson, S. Petitgirard, A. Pakhomova, T. Boffa Ballaran, I. Collings, "High pressure ices and large icy world habitability" ExoOcean workshop, Bern, Switzerland, June 2018.**
- B. Journaux, J.M. Brown, E. Abramson, S. Petitgirard, A. Pakhomova, T. Boffa Ballaran, I. Collings, "High pressure ices equations of state" WaterX workshop, La Madalena, Italy, June 2018.**

(Invited) **B. Journaux**, J. M. Brown, O. Bollengier, E. Abramson, "Deep ice and salty oceans of icy worlds, how high pressures influence their thermodynamics and provide constraints on extraterrestrial habitability". P53H-02, **AGU Fall Meeting**, New Orleans, December 2017.

B. Journaux, M. Montagnat, T. Chauve, F. Barou, A. Tommasi, D. Mainprice, "Heterogeneities During Deformation of Polycrystalline Ice, Recent Advances in Cryo-EBSD Analyses". MR43E-03, **AGU Fall Meeting**, New Orleans, December 2017

B. Journaux, O. Bollengier, J. M. Brown, S. Vance, E. Abramson, "High-pressure water ice and salts thermodynamics: how can physical chemistry constrain the habitability of deep oceans?". **AbSciCon 2017**, Phoenix, AZ, March 2017.

Selected Poster Presentations

(†) U. Jones[§], **B. Journaux**, J. M. Brown, J. A. Hernandez, P. Myint, T. Gordon, G. Tobie, D. Amorim, "A new framework for predicting water-rich exoplanetary interior structures from a new meta-analytical Gibbs energy equation of state for ice VII-X and ice Ih layer thermal model." **AbSciCon 2026**, Madison, WI, USA, May 2026.

(†) I. Rousouli[§], S. D. Vance, C. Woodburn, **B. Journaux**, "On the stability of liquid aqueous solutions in the MgSO₄-NaCl-H₂O system under icy moon conditions." **AbSciCon 2026**, Madison, WI, USA, May 2026.

(†) L. Kafadarian[§], **B. Journaux**, S. D. Vance, J. M. Brown, O. Bollengier, "High-throughput thermodynamic experiments to constrain the physical chemistry and habitability of ocean worlds." **AbSciCon 2026**, Madison, WI, USA, May 2026.

(†) T. Arias[§], **B. Journaux**, "Investigating subsurface temperatures in Permanently Shadowed Regions of the Lunar South Pole." **LPSC 2026**, Woodlands, TX, USA, Mar. 2026.

(Presenting) A. Pakhomova, **B. Journaux**, A. Kurnosov, T. Boffa Ballaran, G. Tobie, M. Hanfland, "Pressure-Induced Methanol Enclathration in Mixed Clathrate Hydrates: Implications for Organic Storage and Methane Production in Ocean Worlds." **LPSC 2026**, Woodlands, TX, USA, Mar. 2026.

(†) E. Patwelka[§], U. Jones[§], **B. Journaux**, R. Barnes, "Simulating Hydrospheres of TRAPPIST-1 h in Search of Liquid Water Layer" **245th Meeting of the American Astronomical Society**, Astronomical Society, Vol. 57, No. 2 e-id 2025n2i362p02, National Harbor, MD, USA, Jan. 2025.

(†) U. Jones[§], **B. Journaux**, J. M. Brown, "Not in a Pickle: On the Water Activity in NaCl(aq) at the Elevated Pressures of Icy Moon Oceans" **AbSciCon**, Providence, RH USA, May 2024.

B. Journaux, J. M. Brown, E. Abramson, S. Vance, "SeaFreeze: Thermodynamics of deep extraterrestrial oceans: effects of pressure, low temperature, composition and equilibrium with solids on habitability" **Conférence Nationale d'Exobiologie**, Marseille, Oct 2021.

(†) W. Dai[§], **B. Journaux**, R. Barnes, "HydroSphere: an open source code to model the structure and thermodynamic equilibria of water dominated planetary shells". P013-0007, **AGU Fall Meeting**, Dec. 2020.

(†) E. Clinton[§], **B. Journaux**, J. M. Brown, "SeaFreeze GUI: A User-Friendly Dive into Water and Ice Thermodynamics of Planetary Abysses". P076-0012, **AGU Fall Meeting**, Dec. 2020.

B. Journaux, J. M. Brown, G. Tobie, O. Bollengier, P. Espinoza, S. Vance, E. Abramson, "SeaFreeze : A modular thermodynamic framework for modeling all solar system planetary hydrospheres, including ice polymorphs and aqueous solutions at high-pressures.", **EPSC-DPS 2019**, Geneva, Switzerland, Sept. 2019.

B. Journaux, J. M. Brown, G. Tobie, O. Bollengier, P. Espinoza, S. Vance, E. Abramson, "Comprehensive thermodynamic representation for planetary oceans and ice shells modeling", **AbSciCon 2019**, Seattle, June 2019.

(Presenting) O. Bollengier, E. Abramson, J. M. Brown, **B. Journaux**, "Experimental Exploration of the Speciation, Solubility and Solid Phases of Carbon in the CO₂-H₂O and CO₂-H₂O-NaCl Systems to 7 GPa and 700 K". DI33B-0036, **AGU Fall Meeting**, Washington DC., December 2018

B. Journaux, "Experimental study of Na-Cl-Mg-SO₄ brines / high pressure ices phase equilibrium: Implication for large icy worlds interior structure and ocean habitability.", **6th ELSI International Symposium**, Tokyo Institute of Technology, Tokyo, Japan, January 2018.

B. Journaux, J. M. Brown, E. Abramson, S. Petitgirard, A. Pakhomova, T. Boffa Ballaran, I. Collings, "High pressure study of water-salt systems, phase equilibria, partitioning, thermodynamic properties and implication for large icy worlds hydrospheres." MR43C-0473, **AGU Fall Meeting**, New Orleans, December 2017.